A PRESCRIPTIVE MODEL FOR STRATEGIC DECISION-MAKING, AN INVENTORY MANAGEMENT DECISION MODEL

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Abstract

In order to reduce overall inventory costs, the analysis focuses on improving inventory management for a manufacturing organization by examining both constant and changing demand scenarios. In Part I, the analysis determines an ideal order quantity of around 677 units utilizing a fixed annual demand of 15,000 units, a cost per unit of $80, a holding cost rate of 18%, and an ordering cost of $220. The overall inventory cost of $9,748.85 is the result of balancing holding costs, which rise with increased inventory levels, and ordering costs, which fall with larger, less frequent orders. The created mathematical model offers a thorough understanding of the dynamics of inventory management by including calculations for annual ordering and holding expenses.

Part II expands the study to a variable demand scenario, employing a triangular probability distribution and a Monte Carlo simulation of 1,000 events to account for demand uncertainty. The simulation predicts a minimum total cost of $9,093.92, with a 95% confidence interval indicating a small range of cost variability. The methodology also forecasts predicted order quantities and annual order volumes, demonstrating how demand unpredictability affects inventory management techniques. The findings highlight the need of adaptive ordering strategies for being cost-effective in the face of demand volatility. The analyses in the study use both Excel and R, demonstrating how these tools may be used in practice to optimize inventory management operations.

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Introduction

For a manufacturing company to be financially stable and run efficiently, inventory management is essential. It is crucial to strike a careful balance between keeping enough inventory on hand to consistently satisfy demand and avoiding the financial strain of having too much inventory.

PART I

Data and Uncontrolled Inputs

Uncontrollable Input	
Annual Demand (D)	15000
Cost per Unit (C)	$80.00
Holding Cost per Rate (R)	18%
Ordering Cost per Order (S)	$220.00
Order Quantity (Q)	677.0032

Table 1 - Data and Uncontrolled Inputs

Inventory management model analysis seeks to decrease overall inventory costs by optimizing order quantity. Based on a 15,000 unit annual demand, a $80 unit cost, $220 ordering cost, and an 18% holding cost rate (i.e., $14.40 unit holding cost annually), the model uses the Excel solver function to determine an optimal order quantity of about 677 units. Ensuring the most cost-effective inventory management technique within the constraints given, this optimal quantity finds a compromise between lowering both the ordering costs, which fall with larger, less frequent orders, and the holding costs, which increase with higher inventory levels.

Model Parameter

Holding Cost per Unit (H)	$14.40
Average Inventory (AI)	338.5016002
Number orders per year (NOY)	22.15646838
Annual Ordering Cost (AOC)	$4,874.42
Annual Holding Cost (AHC)	$4,874.42
Total Inventory Cost (TIC)	$9,748.85

Table 2 - Model Parameter

One of the most important model factors is the holding cost per unit (H), which is $14.40. This is the annual cost of storing one unit of inventory, accounting for charges such as insurance, storage, and opportunity costs. By using this approach, the best order quantity (Q) is determined. Q has a big impact on other model outputs like the Average Inventory (AI) and the Number of Orders per Year (NOY). The average inventory level maintained over time to effectively meet demand while reducing holding costs is represented by the model's computation, which comes out to be roughly 338.5 units on average. Simultaneously, based on the stated annual demand and the computed order quantity, the analysis calculates that roughly 22.16 orders must be issued annually to sustain operations without surplus.

The Annual Ordering Cost (AOC) of $4,874.42, which includes all costs involved with placing orders annually, including those for procurement, shipping, and handling, is directly impacted by the frequency of orders. Similarly, to illustrate the financial consequences of maintaining stock levels, the Annual Holding Cost (AHC), which is likewise computed to be $4,874.42, accounts for the expenses spent from retaining the average inventory over a year. After all of these computations are completed, the Total Inventory Cost (TIC), which includes the yearly costs associated with purchasing and maintaining inventory, comes to $9,748.85.

Mathematical Model

Holding Cost per Unit (H)	H = R x C
Average Inventory (AI)	AI = Q/2
Number of Orders per year (NOY)	NOY = D/Q
Annual Ordering Cost (AOC)	AOC = D/Q x S
Annual Holding Cost (AHC)	AHC = Q/2 x H
Total Inventory Cost (TIC)	TIC = AOC + AHC

Table 3 - Mathematical Model

The holding cost per unit (H), which is the annual cost of storing one unit of inventory, is calculated by multiplying the holding cost rate (R) by the cost per unit (C) in the model. To calculate the average inventory level (AI), which represents the average stock kept over time, half the ideal order quantity (Q) is used. By dividing the annual demand (D) by the order quantity (Q), one can calculate the number of orders per year (NOY), which represents the frequency of order placements required to meet demand. Then, using the order frequency and average inventory level as inputs, the annual ordering cost (AOC) and annual holding cost (AHC) are computed. AOC is the product of NOY and the cost per order (S), while AHC is the product of AI and H. The cumulative cost of inventory management is represented by the total inventory cost (TIC), which is the product of AOC and AHC.

Total Cost vs Order Quantity

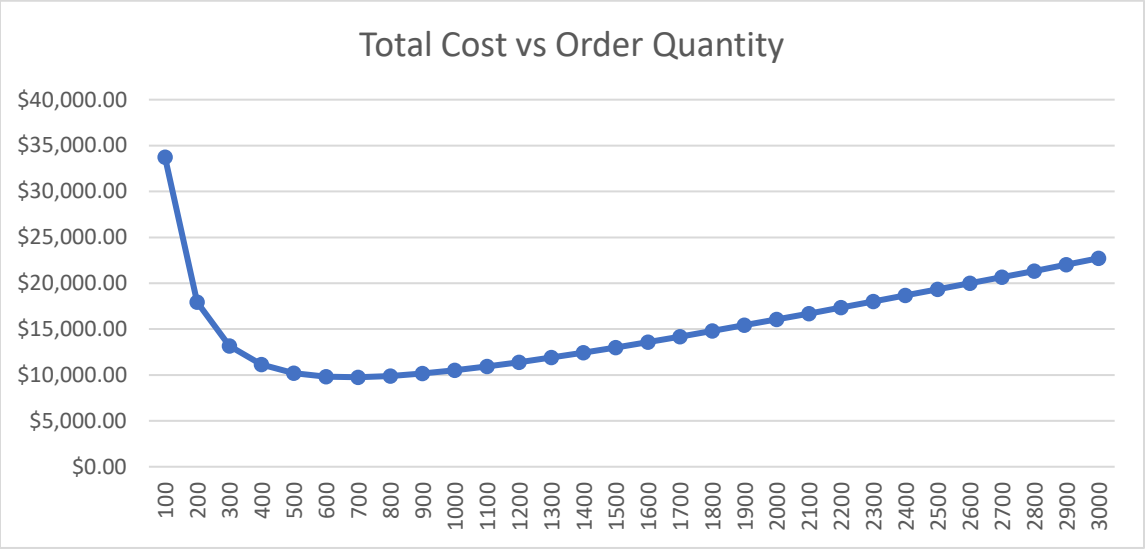

Figure 1 - Total Cost vs Order Quantity

An Economic Order Quantity (EOQ) model is depicted in the graph, which shows the trade-off between order quantities and total inventory costs. Larger orders result in fewer orders overall, which lowers ordering costs. This is demonstrated by the graph's initial sharp fall in costs as order size rises. The EOQ is represented by the lowest point on the curve, which is the point at which total costs are at their lowest and an ideal balance between the costs of ordering and keeping inventory is reached. The curve progressively increases as the order quantity exceeds the EOQ, signifying increased holding costs as a result of bigger amounts of inventory being held. Businesses can reduce their overall inventory management expenses by using this visualization to identify the most economical order quantity.

Solver and Optimize Function

Order Quantity (Q)	677.0032004
Total Inventory Cost (TIC)	$9,748.85

Table 4 - result of Solver and Optimize Function

The solver and optimize function was used to find the order quantity that minimizes the total inventory cost, given a certain demand. According to the results, the minimum total cost of $9,748.846 is achieved at an order quantity of approximately 677 units. This is the point at which the combined cost of ordering and holding inventory is lowest.

What-If Analyses by Using Two-Way Tables

Different order quantities are represented in the leftmost column, and progressively increasing demand is shown in the top row. Next, for each order quantity and demand combination, the total cost is shown in each cell of the table. The table, with green denoting lower costs and red greater expenses, represents various cost ranges. What-if analysis is run using this data to determine how sensitive changes in these two parameters (order quantity and Cost) will be to the overall cost. For example, you can find the order quantity that, for a given demand, yields the lowest total cost. On the other hand, you may observe how variations in demand impact the overall cost for a set order quantity.

Summary of Part I

The goal of the initial part of the model analysis is to find the ideal order quantity, which is based on factors like holding cost rate, cost per unit, annual demand, and ordering cost. This will reduce the overall inventory costs. According to the analysis, an order quantity of roughly 677 units is ideal for a demand of 15,000 units annually, a cost per unit of $80, an ordering cost of $220 per order, and an 18% holding cost rate, which translates to an annual holding cost per unit of $14.40. This strategic amount strikes a compromise between holding costs—which rise with higher inventories—and ordering costs, which fall with fewer, larger purchases. Based on the model's calculations, an average inventory of 338.5 units is needed annually, or roughly 22.16 orders. This results in $4,874.42 yearly ordering costs and $4,874.42 yearly holding costs, for a total of $9,748.85 in inventory costs. The outputs of the mathematical model are based on simple calculations: the number of orders per year is determined by dividing the annual demand by the order quantity; holding cost per unit is calculated as the product of holding cost rate and cost per unit; average inventory is calculated as half the order quantity; and the

total inventory cost is the sum of the annual ordering and holding costs. Better inventory decisions are made possible by the two-way tables, which make it easier to do what-if analysis to determine how sensitive the overall cost is to variations in order quantity and demand.

In conclusion, by determining the ideal order quantity and understanding its impact on other aspects of inventory management, such as average inventory level, the number of orders per year, and total costs, businesses can effectively optimize their inventory costs. The visual representation of total costs against order quantities, depicted in the provided figures, illustrates the trade-off between these elements and aids in identifying the most cost-efficient strategies for inventory management.

PART II

We explore a scenario where the annual demand is variable rather than constant in this extended analysis of inventory management. This scenario is based on a triangular probability distribution with a most likely value or mode of 15,000 units, an upper limit of 17,000 units, and a lower limit of 13,000 units. This method allows for the inclusion of uncertainty in our model and depicts a more realistic scenario where demand swings within a specific range. All other metrics, including cost per unit, ordering cost, and holding cost rate, stay the same as they were in the original scenario, even with this variation in demand.

Monte Carlo Simulation for Inventory Management

The Monte Carlo simulation using a triangular distribution of annual demand presents an insightful analysis for inventory management under uncertain demand conditions. Through the modelling of 1,000 demand scenarios, with a mode of 15,000 units and a range of 13,000 to 17,000 units, the best order quantities and accompanying total inventory costs were determined for each scenario. The different ideal order quantities point to the necessity for adaptive ordering strategies to preserve cost-effectiveness, and the diversity in the simulated total costs emphasizes the direct impact of demand uncertainty on inventory charges.

Expected Minimum Total Cost with 95% Confidence Interval

The most effective position for controlling inventory costs under the specified demand distribution was found at $9,093.92, which was the simulation's lowest total cost. The simulated total cost outcomes were concentrated around this mean value within the range of the simulated demand, as indicated by the simulated total cost mean of roughly $9,752.90 and standard deviation of $270.76.
The estimated minimum total cost was found to have a high degree of confidence due to its relatively narrow range, as indicated by the 95% confidence interval, which falls between $9,736.10 and $9,769.70.

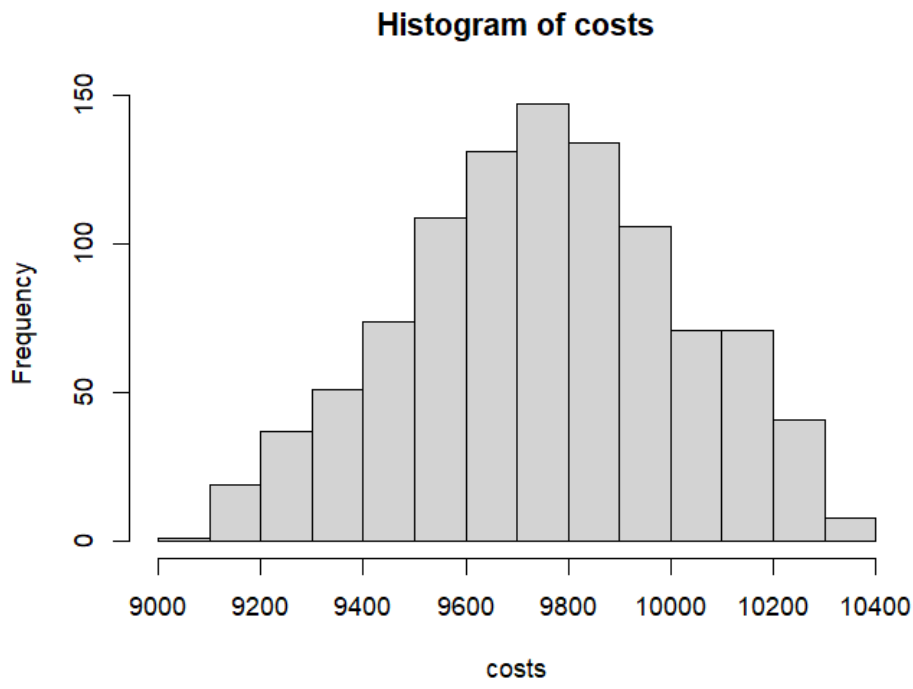

Figure 2 - Histogram of Cost

The histogram represents the distribution of total inventory costs based on 1,000 simulations, with a mean cost of approximately $9,752. The distribution is concentrated mostly around the $9,700 to $10,000 range, showing the most common total cost outcomes from the simulation. The histogram suggests a roughly normal distribution with a slight right skew, implying that while the majority of the costs are centred around the mean, there are a few instances of higher costs. The tails of the distribution taper off, indicating fewer occurrences of extremely low or high costs.

Validity and Goodness of Fit for Minimum Total Cost

The Chi-squared test revealed a significant difference between the observed frequencies of simulated total costs and what would be predicted if the costs were normally distributed, with a Chi-squared statistic of 26.505 and 13 degrees of freedom, yielding a p-value of 0.01453. This suggests rejecting the null hypothesis that the data follows a normal distribution. The calculated cumulative probability of 0.9854681 supports this conclusion, indicating that the observed data distribution is significantly different from a normal distribution at the customary 5% significance level.

Expected Order Quantity with 95% Confidence Interval

The simulated data was used to calculate the mean order quantity, which is roughly 677 units. The standard deviation shows how different the order quantities are from and around this mean. The range of approximately 676.12 to 678.45 is the 95% confidence interval, which gives an estimate of the interval with 95% certainty within which the true mean order amount is estimated to reside.

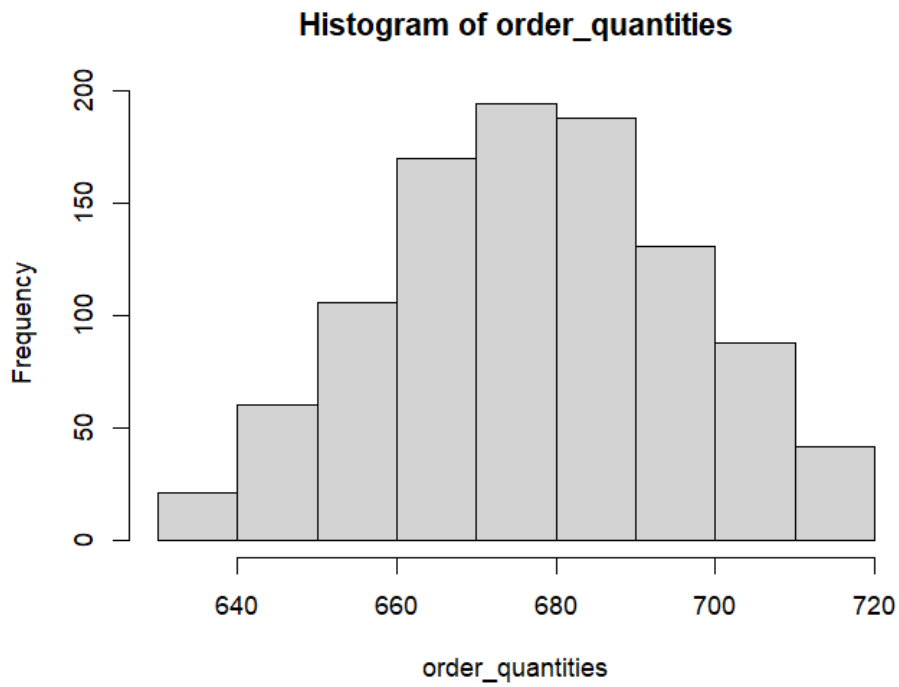

Figure 3 - Histogram of Order Quantities

A highly symmetrical distribution centred around the 670s is seen in the histogram of order quantities from a simulation, indicating a normal distribution of the data. With a narrow 95% confidence interval that spans from roughly 676 to 678 units, the mean order amount is roughly 677 units, suggesting precision around the mean estimate. The simulated order quantities are mostly near the mean, with fewer quantities as we approach away from the distribution centre, as evidenced by the largest frequency falling between 670 and 680. The histogram's bell-shaped curve, which represents consistency in the simulation output, supports the notion that order quantities are normally distributed around the computed mean.

Validity and Goodness of Fit for Order Quantities

The Chi-squared test on the observed order quantities yielded a statistic of 13.342 with 8 degrees of freedom and a p-value of 0.1006. This p-value indicates that the distribution of order amounts is not statistically different from a normal distribution, as it is greater than the conventional alpha level of 0.05, suggesting no rejection of the null hypothesis. The results show that the simulated order amounts are normally distributed with a mean of 676.7851 units and a standard deviation of 0.5838.

Expected Annual Number of Orders with 95% Confidence Interval

The analysis computes the average number of orders per year using various simulated requests, yielding a mean of about 1.0015 orders per year with a standard deviation of 0.0559. The 95% confidence interval for the expected number of orders per year is between 0.9980 and 1.0049, indicating that the true mean of orders per year sits within this range provided the identical conditions are duplicated over numerous samples.

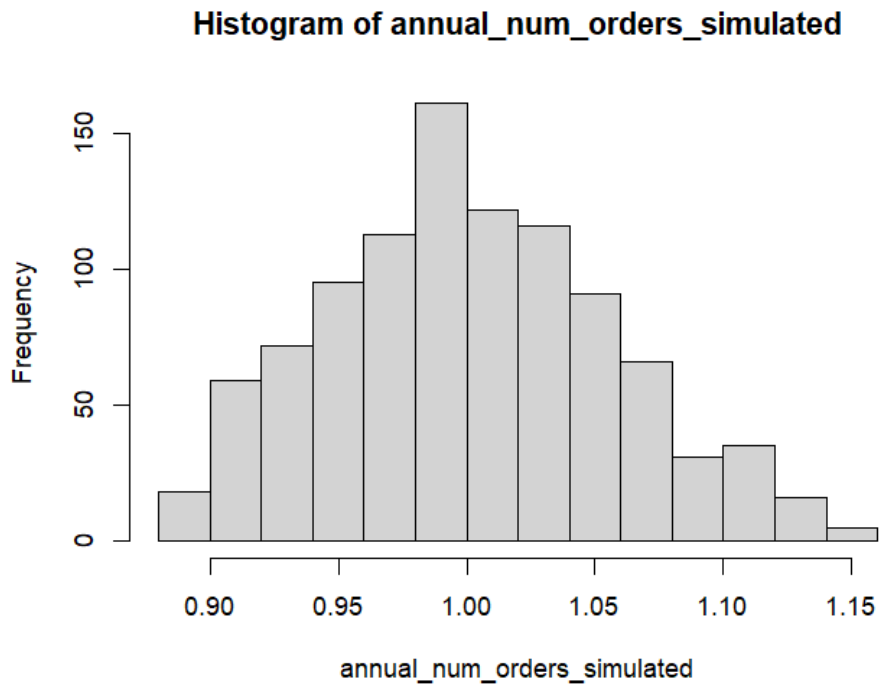

Figure 4 - Histogram of Annual Number of Orders

A histogram of the simulated annual number of orders reveals that the data is concentrated around the mean with the highest frequency at the center of the distribution, indicating a likely normal distribution. The breaks of the histogram are evenly distributed between 0.88 and 1.16, covering the range of observed orders per year. The density shows the frequency of the simulated data within each bin, with the central bins around the mean of 1.00 having the highest density, illustrating that most of the annual order number simulations are close to 1 order per year, which aligns with the mean annual number of orders.

Validity and Goodness of Fit for Annual Number of Orders

The simulation results, subjected to a Chi-squared test, show that the mean annual number of orders is just over 1 with a very tight confidence interval, indicating consistency in the ordering pattern. However, with a p-value of 0.0062 indicating a significant difference from a normal distribution, the results suggest that the annual number of orders may be influenced by factors that cause deviation from normality or perhaps the sample size is not adequately capturing the distribution's true characteristics.

Conclusion

The inventory management analysis indicates that, at a $80 cost per unit and an 18% holding cost rate, the ideal order quantity for a company with a steady yearly demand of 15,000 units is about 677 units. By striking a balance between holding costs, which rise with increased inventory levels, and ordering costs, which are lowered by larger and less frequent orders, this ideal position is reached. A best-case scenario with the lowest projected total inventory cost of $9,093.92 is suggested by the simulation conducted under conditions of fluctuating demand using a Monte Carlo technique and a triangular distribution.

The Chi-squared test, however, shows notable departures from the normal distribution for the annual number of orders as well as the overall expenses, suggesting the presence of extraneous variables or a small sample size that may not adequately capture the true distribution.

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